From waste rock to clean energy: real options valuation of renewable energy systems integrating silicate rock by-products generated by the mining sector in the Colombian Caribbean region

Hugo Hernández Palma[1] | **Anderson Isaías Nieto Granados**[1]

[1] *Facultad de Ingeniería, Corporación Universitaria Iberoamericana, Bogotá, Colombia*

1. INTRODUCTION

The Colombian Caribbean region has significant mining activity that generates large volumes of silicate-rich waste rock. This project evaluates the techno-economic feasibility of integrating these by-products into renewable energy systems through a real options approach (LSMC), considering uncertainties in energy prices, material availability and investment decisions.

MINING ZONES – COLOMBIAN CARIBBEAN



Focus: Silicate-rich waste rock from coal and industrial mineral operations in La Guajira and Cesar.

2. OBJECTIVES

- **Assess** the potential of silicate rock by-products as inputs for renewable energy infrastructure and storage materials.
- Model the integration of waste rock recovery with renewable energy systems (solar PV, BESS, microgrids).
- Apply Least-Squares Monte Carlo (LSMC) real options valuation to optimize investment decisions under uncertainty.
- Provide decision-support insights for sustainable mining and clean energy transition in the Caribbean region.

3. METHODOLOGY – SYSTEMS, AUTOMATION & LSMC

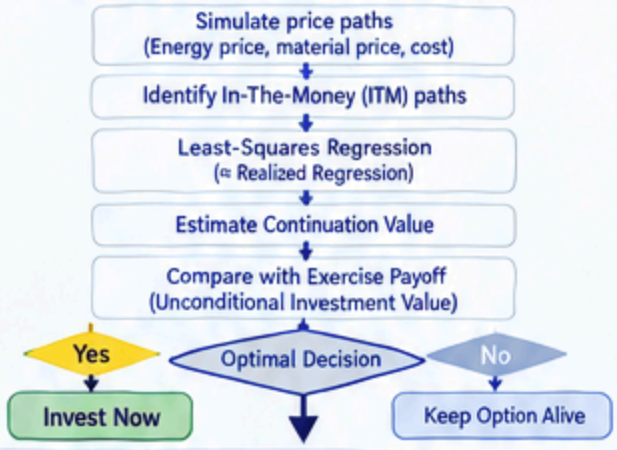
1 Automated Waste Rock Recovery Unit



2 Integration into Renewable Energy Systems



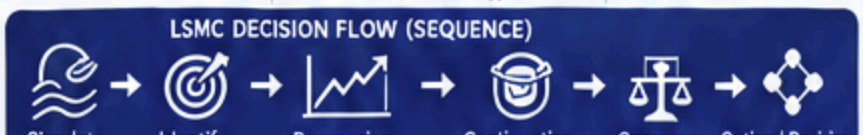
3 Real Options Valuation Engine (LSMC)



4 IoT Monitoring & Automated Control Layer



LSMC DECISION FLOW (SEQUENCE)



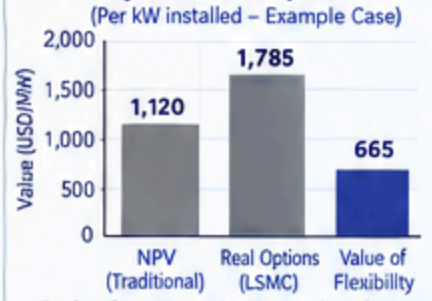
4. KEY RESULTS

Option Value vs. Energy Price
(LSMC Real Option Valuation)



Higher energy prices increase the value of the option to invest. Flexibility to defer adds significant value.

Project Value Comparison
(Per kW installed – Example Case)



Real options approach captures the value of managerial flexibility, improving project attractiveness.

5. CONCLUSIONS

- Waste rock from mining can be transformed into valuable inputs for renewable energy systems, promoting circular economy principles.
- LSMC real options valuation provides a robust framework to make flexible investment decisions under uncertainty.
- Integration of automation, IoT, and advanced analytics enhances system performance, reduces risks, and maximizes value.
- This approach supports the sustainable energy transition and strengthens the competitiveness of the mining sector in the Colombian Caribbean.

IMPLICATIONS

- **Policy:** Incentives for by-product valorization and clean energy projects.
- **Industry:** New revenue streams from waste rock and lower carbon footprint.
- **Society:** Cleaner energy access, local development, and environmental stewardship.
- **Decision Tool:** LSMC enables better timing and sizing of investments under uncertainty.

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